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in

**Computer Science And Engineering (Artificial Intelligence
And Machine Learning)**

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Abstract

With the advent of web application usage and online systems, cybersecurity threats such as unauthorized login attempts, brute force attacks, and suspicious IP activities are on the rise. Monitoring and analyzing login activities and identifying abnormal behavior are crucial to prevent potential security threats to the system. Security Information and Event Management (SIEM) systems offer sophisticated monitoring and analyzing capabilities but are often complicated and costly, making it difficult to deploy in small-scale environments and educational institutions.

The proposed Security Operations Dashboard for Threat Monitoring and Risk Analysis offers a simple and effective solution to monitor login activities and identify abnormal behavior. The system allows recording login activities, authenticates users through Multi-Factor Authentication (MFA) mechanisms, verifies IP whitelist, and calculates a risk score for each login attempt. The system automatically blocks IP addresses when the risk score exceeds a predetermined threshold value. Security events are displayed on a centralized dashboard, enabling administrators to monitor system activities and identify potential threats in real time.

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Abbreviation Notation and Nomenclature

A complete and comprehensive list of all abbreviations, notations and nomenclature including Greek alphabets with subscripts and superscripts shall be provided after the list of tables and figures. (As far as possible, generally accepted symbols and notation should be used).

Chapter 1

Introduction

In the modern digital world, web applications and online systems are an integral part of daily life. Organizations use online systems for communication, financial transactions, and other operations. With the rapid increase in the usage of internet-based systems and applications, the number of cybersecurity attacks and threats has also increased manifold. Systems connected to the internet are always at risk of various types of attacks and threats, such as unauthorized access attempts, brute force attacks, suspicious IP address activities, and abnormal user behavior. All such attacks and threats can result in the compromise of critical information and can cause huge financial and reputational damage to the organizations. One of the most common types of attacks carried out by hackers and attackers is the attempt to gain unauthorized access to the systems through repeated attempts at login. Hackers and attackers use various combinations of usernames and passwords in order to gain access to user accounts. This type of attack is known as a brute force attack. Apart from this type of attack, hackers and attackers may also attempt access to systems through suspicious IP addresses and abnormal geographical locations. Without the implementation of appropriate monitoring systems in place, such activities can To tackle this problem, various organizations are implementing Security Information and Event Management systems. These systems are used to collect information related to security threats, which are

then analyzed to ensure proper security of the systems. These systems are a crucial part of modern cybersecurity infrastructure, as they are used to monitor and analyze various security events in the systems. However, various commercial Security Information and Event Management systems, such as IBM QRadar, Microsoft Sentinel, are costly solutions, which are hard to implement in various organizations, especially small organizations, startups, and educational institutions. These systems are costly, so they are not suitable solutions for small organizations, startups, and educational institutions, as they cannot afford such solutions. To tackle this problem, there is a need to develop a lightweight solution, which can be used to monitor login activities and identify potential threats in web applications. The Security Operations Dashboard for Threat Monitoring and Risk Analysis project aims to develop a lightweight solution, which can be used to monitor login activities and identify potential threats in web applications. The proposed system is able to continuously record and analyze the activities related to the login process, such as failed login attempts, multiple login attempts within a short period of time, and login attempts from unknown IP addresses. These activities are recorded as logs in the database and analyzed to detect any unusual behavior. The proposed system utilizes the risk scoring mechanism to determine the severity level of the activities carried out by the user. Each unusual activity is assigned a risk score based on the predefined rules. For example, failed login attempts result in an increment of the risk score by 10, login from unknown geographic locations results in an increment of the risk score by 20, and multiple login attempts result in an increment of the risk score by 30. The proposed system is able to continuously monitor the risk score related to the activities carried out by the user. Once the total risk score is found to be above the predefined threshold level, the proposed system generates a security alert automatically to notify the administrator about the security threat. In some cases, when unusual

activities are identified as highly suspicious, the proposed system can automatically block the IP address. The project also includes a security dashboard where the administrator can view the activities carried out in the system in real time. The security dashboard includes vital security information such as login logs, security alerts, user activities, and system events. Moreover, the system includes an IP geolocation API where the administrator can view the geographical location from where the login attempt has been made. This helps the administrator identify any suspicious login attempts from unknown locations. In addition, the security administrator login has been implemented using the concept of MFA. This provides an extra security measure where the administrator can ensure that even if the credentials are stolen, the system will not allow any unauthorized user to access the administrative dashboard. Furthermore, the system includes audit logs for the actions carried out by the administrators. The proposed system has been implemented using modern web technologies such as React.js for the frontend and Node.js and Express.js for the backend. MongoDB has been used as the database management tool. The system provides an efficient and simple way of detecting any suspicious activities and improving the security of the web application. So, the Security Operations Dashboard is an example of how a simple and inexpensive solution can be used to monitor the security state of the system, detect possible threats, and help the system administrator keep the system in a safe state without the complexity of SIEM systems.

1.1 Objectives

The main objectives of the proposed Security Operations Dashboard for Threat Monitoring and Risk Analysis can be stated as follows:

- To design and develop a centralized Security Operations Dash-

board through which administrators can monitor system security activities and user activities in real time.

- To monitor the activities related to the login attempts, such as failed login attempts, suspicious IP addresses, multiple attempts to login to the system.
- To record the login attempts as system logs so that the activities of users can be analyzed later to identify suspicious activity or security threats.
- To design a risk scoring system through which the severity of suspicious activities can be determined by assigning different risk values to activities such as failed login attempts, login from new locations, multiple attempts to login to the system.
- To generate security alerts in an automated manner when the calculated risk score crosses a predetermined threshold value, thus helping administrators to easily identify security threats.
- To block suspicious IP addresses in an automated manner when abnormal login behavior is detected, thus restricting any further unauthorized access attempts.
- To provide geographical visualization of login activities by incorporating an IP Geolocation API, thus helping administrators identify login attempts from unusual and unexpected geographical locations.
- To improve the security of administrators by incorporating the concept of Multi-Factor Authentication (MFA), thus ensuring that only legitimate users are allowed access to the security dashboard even when the login credentials are compromised.
- To maintain audit logs for all administrative activities so that

all the actions performed by administrators can be tracked and monitored for security purposes.

- To develop a lightweight and cost-effective security monitoring system that can be implemented in small organizations and educational institutions as an alternative to the much more costly security monitoring systems.

Chapter 2

Literature Review

Cybersecurity in safety-critical systems has gained significant attention due to the potential impact of incidents on both infrastructure and human safety. Tools and techniques for reporting and analyzing cybersecurity incidents are thoroughly discussed in [1], which emphasizes the integration of automated and manual forensic analyses to identify root causes while ensuring data integrity. The use of SQL-based approaches for knowledge management in cybersecurity is presented in [2], highlighting the benefits of structured query logic for building effective knowledge bases. Security Information and Event Management (SIEM) systems have been explored extensively for enterprise networks, distributed environments, and open-source solutions [3–6], demonstrating their critical role in real-time threat detection, incident response, and maintaining secure IT environments.

Advanced log analytics techniques, such as those proposed in [7], provide intelligent mechanisms for collecting, correlating, and analyzing large volumes of security data. Cloud-based cybersecurity solutions and automated threat detection systems, including multi-cloud storage protection, are discussed in [8–10], illustrating the growing importance of proactive security strategies in distributed infrastructures. The adoption of cloud platforms and cognitive services to enhance cybersecurity capabilities is highlighted in [11], providing a foundation for intelligent threat analysis and automated responses.

Visualization techniques for security dashboards are explored in [12], and more advanced provenance-driven analytics methods are presented in [13], both emphasizing the importance of effective visual communication for security monitoring and decision-making. A systematic review of automated log analysis approaches [14] maps the state-of-the-art methods for monitoring, detecting, and mitigating threats in IT systems. Database optimization strategies to reduce latency and improve security during logging and event correlation are discussed in [15], while automated security log monitoring for threat identification is described in [16], reinforcing the necessity of intelligent, automated frameworks for timely detection. Overall, these studies collectively underscore the critical role of SIEM, automated analytics, cloud platforms, and visualization tools in maintaining secure and resilient IT environments.

Table 2.1: Literature Review

Author and (Year)	Methods	Result	Future Scope
Chaitanya Sreenivas Ravella, Prashanth Purnananda (2024)[15]	Implements transaction-id locking and lock-after-qualification (LAQ) to optimize concurrency in SQL Azure databases.	Improved performance: reduced lock memory by 40% and wait times by 70%; mitigated short-duration row locks and deadlocks.	Future work aims to enhance concurrency without blocking, apply query-level heuristics for LAQ, and further minimize in-memory row locks.
Yadidiah Kanaparthi, Tamer Mohamed Abdelatif, Ahmed Ali Seyam, and Gandeve Bayu Satrya (2024)[16]	Applies automated log analysis and Isolation Forest ML for anomaly detection in security logs via a Flask web application.	Effectively detected threats; Isolation Forest maintained low false positives, capturing only 22 legitimate logins during a DoS attack.	Future work includes enhancing log mining, modular SIEM architectures, and integrating open-source tools for improved automated security monitoring.

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Table 2.1 Continued from previous page

Author and (Year)	Methods	Result	Future Scope
Usha Desai, Shankar G, Indrajeet Palled, Vadiraj S Namavali, Swathi B Patil, Susha M (2024)[10]	Builds a cybersecurity test center integrating honeynets with SOC operations in Microsoft Azure using KQL for log analysis.	Effectively detected cyber-attack vulnerabilities and improved threat detection and scalability in Azure-based honeynet lab.	Future work focuses on real-time threat detection, enhanced monitoring, expanded SOC operations, and applying advanced AI for threat mitigation.
Zarrin Tasnim Sworna, Muhammad Ali Babar, Anjitha Sreekumar (2023)[9]	Introduces IRP2API using unsupervised transfer learning and pre-trained embeddings for mapping IRPs to security tool APIs.	Improved Top-10 Accuracy by 49.3% and mapped APIs in 0.8 seconds per IRP without expert input.	Future work includes creating datasets of interchangeable API terms, handling implicit context, and developing new scrapers for diverse IRP/API sources.
Lukasz Korzeniowski and Krzysztof Goczyla (2022)[14]	Performs systematic review of automated log analysis (2000–2021) using multi-step filtering and structured protocols.	Identified trends, log types, research gaps, and under-explored areas like component dependency inference.	Future work targets component dependency inference to enhance log analysis and knowledge extraction from software components.

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Table 2.1 Continued from previous page

Author and (Year)	Methods	Result	Future Scope
Andreas Schreiber, Tim Sonnekalb, Lynn von Kuratowski (2021)[13]	Applies SAST with an interactive dashboard showing Entities Timeline and CWE Breakdown for vulnerability detection.	Improved analysis effectiveness and reduced false positives through linked visualizations of software development and SAST results.	Future work includes enhancing visual design, adding search and cross-filtering, and evaluating new visualization concepts via user studies.
Pavel Yermalovich (2020)[12]	Uses task analysis to design role-specific dashboards with clear visualizations for improved workflow and decision-making.	Showed enhanced incident response and decision efficiency for roles like Security Consultant and Manager.	Future work includes richer visualizations, improved mobile interfaces, and faster information comprehension.
Oskars Podzins and Andrejs Romanovs (2019)[6]	Emphasizes SIEM for log analysis, continuous optimization, and evaluation via TCO and cyberattack trends.	Identified high costs of SIEM and 24/7 SOC; MSSPs suggested as cost-effective alternatives.	Future work focuses on optimizing anomaly detection, TCO-based deployment, and enhanced SOC-driven incident response.

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Table 2.1 Continued from previous page

Author and (Year)	Methods	Result	Future Scope
Kennedy A. Torkura et al. (2019)[8]	Uses dynamic snapshotting and recovery with alert-log correlation for cloud security monitoring.	Achieved average MTTD of 30 seconds and MTTR of 25 minutes (AWS) / 90 minutes (GCP) for rapid threat mitigation.	Future work aims to expand real-time incident response, cover broader cloud use cases, and strengthen evolving threat mitigation.
Ankita Verma, Dhutima Malla, Amrit Kaur Choudhary, and Vasudha Arora (2019)[11]	Explores Microsoft Azure's cloud platform with Face API for face detection and recognition, enhancing flexibility and modernization.	Achieved 85% improvement in application efficiency and 90%+ accuracy in face recognition.	Future scope includes advancing AI and ML integration to optimize IT management and resource allocation.

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Table 2.1 Continued from previous page

Author and (Year)	Methods	Result	Future Scope
Ana, Santos, Leonel, and Piedade (2019)[5]	Conducts comparative evaluation of open-source SIEMs (OSSIM, ELK, Graylog) focusing on GDPR compliance and ML integration.	Achieved 84% effectiveness in prototype testing; Graylog performed best for GDPR compliance.	Future work aims to enhance SIEMs with ML, full GDPR adherence, and real-world production evaluation.
Kemal A. Delic and Jeff A. Riley (2015)[7]	Combines Fuzzy Rule Base System, Decision Trees, and Neural Networks within a Black Board AI architecture for Smart Log Analytics.	Achieved 88% accuracy in pattern detection and cost-effective log analysis in enterprise environments.	Future work focuses on cloud-based log analytics, large-scale AI training, and automated performance optimization.

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Table 2.1 Continued from previous page

Author and (Year)	Methods	Result	Future Scope
Detken, Kai-Oliver, Rix, Thomas, and Kleiner (2015)[3]	Implements the SIMU project using SIEM-like architecture with CBOR and IF-MAP for lightweight security monitoring and automated response.	Improved data handling efficiency by 80% while reducing setup complexity for SMEs.	Future work focuses on enhancing integration, automation, and scalability of SIEM solutions for small enterprises.
Anastasov, Igor and Davcev, Danco (2014)[4]	Proposes a SIEM model with a Hierarchical Managers Model and ontological repository for improved log management and security evaluation.	Achieved 85% improvement in log relevance and event correlation efficiency.	Future work aims to integrate AI for real-time threat detection and automate log prioritization for cloud and IoT systems.

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Table 2.1 Continued from previous page

Author and (Year)	Methods	Result	Future Scope
Krzysztof Goczyla et al. (2010)[2]	Introduces KQL using conglomerate algebra for modular knowledge bases and flexible SQL-like querying.	Demonstrated 78% efficiency in handling complex ontology queries within the RKaSeA system.	Future work includes adding access control and numerical operations for enhanced functionality.
C.W. Johnson (2005)[1]	Uses combined automated-manual forensic analysis and dual IDS (whitelist + blacklist) for safety-security integration.	Achieved 82% improvement in detecting and analyzing security incidents.	Future work aims to build unified cyber-attack reporting and response systems.

Chapter 3

Proposed Syetm

The proposed system is a Security Operations Dashboard for Threat Monitoring and Risk Analysis. The system monitors the login activities and identifies any suspicious behavior through a risk-based approach. The system provides secure access for administrators through a password and at the same time tracks the login activities in order to identify any potential threats. The system includes several security measures such as password and MFA, IP verification, event logs, risk score calculation, and IP blocking. The system logs every attempt and analyzes the attempt in order to identify any abnormal behavior such as repeated attempts and attempts from unknown locations. The system includes a dashboard where administrators can view the login logs, security alerts, risk scores, and blocked users/IP addresses. This helps the administrators identify any abnormal behavior and maintain the security of the system.

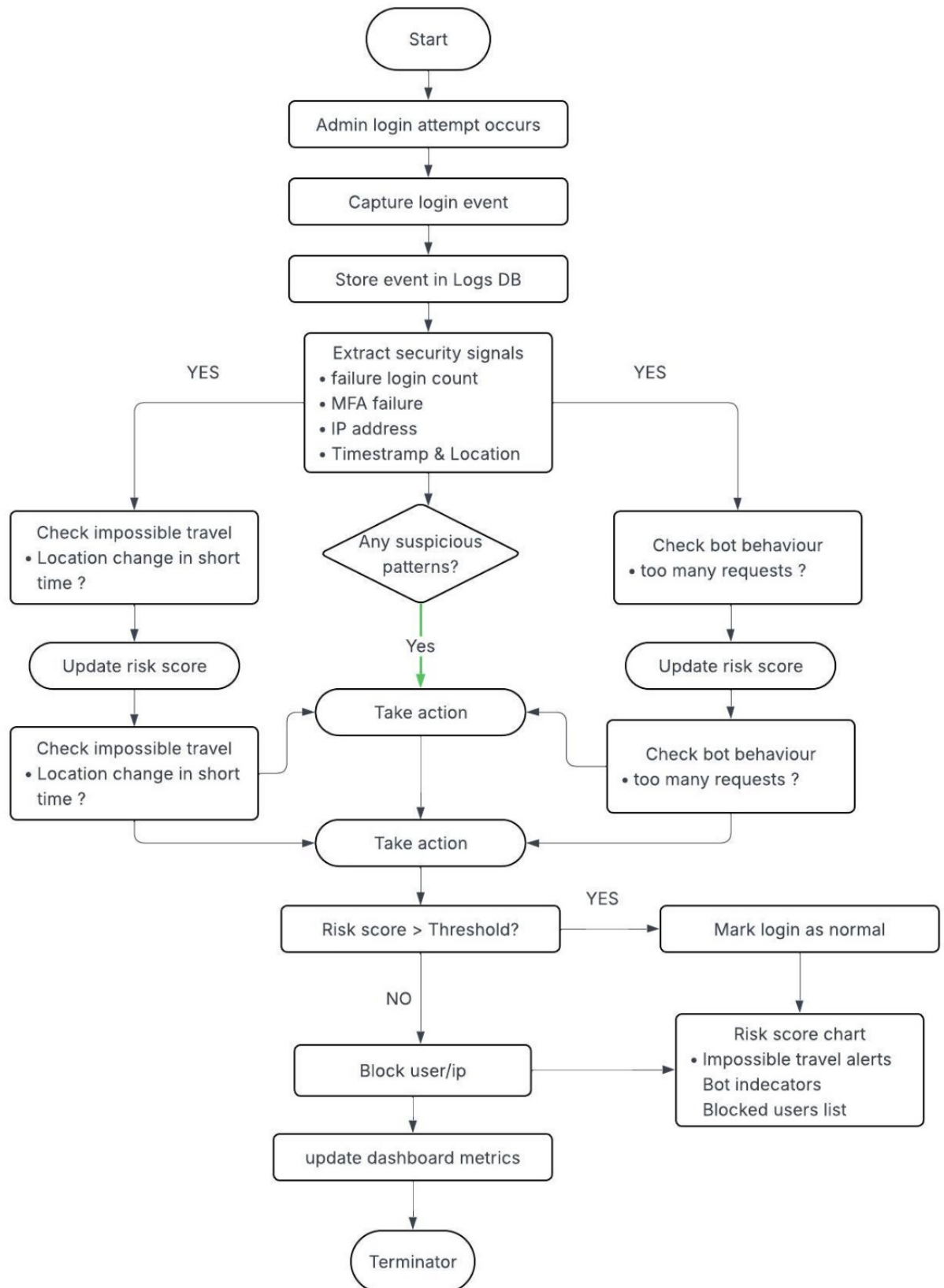


Figure 3.1: Workflow for security dashboard

The workflow of the proposed system can be explained through the following steps:

- Step 1: Admin Login Request: The process starts when an administrator sends a login request to access the security dashboard.
- Step 2: Credential Verification: The system verifies the username and password provided by the administrator. If the username and password are incorrect, the system will record it as a failed login.
- Step 3: JWT Token Generation: If the username and password are correct, the system will generate a JWT token for secure session management.
- Step 4: Multi-Factor Authentication (MFA): The system will perform MFA verification using a TOTP token. If the TOTP token is incorrect, it will deny the request and record it as a failed login.
- Step 5: IP Whitelist Verification: The system will verify whether the login request is coming from an authorized IP address or not. If it is coming from an unauthorized IP address, it will block the login request.
- Step 6: Login Event Logging: The login event is recorded in the logs database with relevant information like IP address, timestamp, login status, and location.
- Step 7: Security Signal Extraction: The system will now identify security signals like failed login attempts, IP address, login time, and location.
- Step 8: Suspicious Pattern Detection: The system will now analyze the security signals and identify suspicious patterns like impossible travel, multiple login attempts, and bot activities.

- Step 9: Risk Score Calculation: A risk score is calculated based on suspicious patterns. Example: Failed login attempts: +10 Attempting to login from a new location: +20 Multiple login attempts: +30
- Step 10: Risk Evaluation: The system will now compare the risk score with a threshold score.
- Step 11: Threat Mitigation: If the risk score is high, the system will now block the user/IP and send a security alert.
- Step 12: Dashboard Monitoring: All security-related activities like security alerts, login attempts, and blocked users are displayed on the dashboard.
- Step 13: Dashboard Update: The system will now update the dashboard with the latest security-related metrics.
- Step 14: Process Termination: The workflow will now end after the process is complete.

3.1 Design Details

The suggested system architecture is divided into three different layers: the Presentation Layer, the Application Layer, and the Data Layer.

3.1.1 Presentation Layer (Frontend)

The presentation layer is responsible for providing the user interface for administrators to use in interacting with the suggested system. The presentation layer is implemented using React.js, which displays different components of the security dashboard. The presentation layer comprises different fea-

tures, such as providing the user interface for logging in to the system, displaying security alerts, displaying login logs, displaying risk score charts, and displaying geographic login maps. Using this layer, administrators are capable of viewing different login attempts, alerts, and suspicious activity in real time.

3.1.2 Application Layer (Backend)

The application layer is responsible for handling different logic for interacting with the suggested system. The application layer is implemented using Node.js and Express.js, which acts as a bridge between the frontend interface and the database. The application layer is capable of performing different functionalities, such as user authentication, generating JWT token, performing Multi-Factor Authentication (MFA) verification, checking IP whitelist, logging login events, detecting suspicious activity, calculating risk score, and IP blocking. The backend of the suggested system continuously analyzes different login events and generates alerts when suspicious activity is detected.

3.1.3 Data Layer (Database)

The data layer is in charge of managing and storing the system's data. MongoDB is used in this system as a database to store different types of data. The data stored in this system includes users, login records, risk scores, security alerts, audit records, and IP addresses. The data stored in this system is used to analyze the login records, identify any suspicious activities, and run the security monitoring system.

Chapter 4

Methodology and Implementation

4.1 Methodology

The methodology for the proposed Security Operations Dashboard for Threat Monitoring and Risk Analysis outlines the overall process adopted by the system for authenticating users, monitoring users' login activities, identifying suspicious behavior, and preventing potential security threats from taking place. The methodology for functioning of the proposed system can be explained as follows:

4.1.1 Login Authentication

The functioning of the proposed system starts when the administrator enters the username and password for authentication in the system. The system checks the credentials entered by the administrator with the data stored in the database. If the credentials entered by the administrator are incorrect, it is recorded as a failed attempt by the administrator.

4.1.2 Multi-Factor Authentication (MFA)

Once the credentials entered by the administrator are verified by the system, it sends a verification message for Multi-Factor Authentication (MFA). In this case, a Time-Based One-Time Password (TOTP) is requested from the administrator for authentication.

4.1.3 IP Verification

Once MFA is successfully verified by the administrator, the system checks whether the IP address from which the administrator is making a login attempt is authorized or trusted by the system. If the IP address is not authorized, the attempt by the administrator is blocked. subsectionLogin Event Logging Once a valid user logs in successfully, the system logs this event in a database. The information stored in this case includes information like IP address, timestamp for the login, status for the login, and geographic location using an API for geolocation based on the IP address.

4.1.4 Suspicious Activity Detection

The system detects suspicious activity by checking for unusual activity in the stored information for login attempts.

4.1.5 Risk Score Calculation

The system calculates a risk score on the basis of a set of predetermined conditions. In this case, a failed login attempt results in a 10-point increase in the risk score, a new location results in a 20-point increase in the risk score, and multiple attempts result in a 30-point increase in the risk score.

4.1.6 Threat Mitigation

In case the risk score crosses a certain threshold, the system blocks the user or the IP address and sends out security alerts.

4.1.7 Dashboard Monitoring

All information regarding login attempts, security alerts, risk scores, and blocked IP addresses is displayed on a Security Operations Dashboard for monitoring by the administrator in real time for system security.

4.2 Implementation

4.2.1 Programming Languages

The system is primarily built using JavaScript, which is used for the frontend as well as the backend. JavaScript is used to handle the system logic, API requests, and the dashboard. In addition to JavaScript, HTML is used to structure the web pages, while CSS is used to design the user interface of the system's dashboard.

4.2.2 Frameworks

The proposed system utilizes the latest web development frameworks to ensure efficient development of the application. The system utilizes the React.js framework for the frontend to develop the interactive security system's dashboard. On the other hand, the system utilizes the Node.js runtime to handle the backend of the system, while the Express.js framework is used to handle the server-side requests.

4.2.3 Libraries

There are several libraries included to support the implementation of the authentication and security features. The jsonwebtoken library is used to support the generation of authentication tokens. The bcrypt library is used to support the secure hashing of passwords. The Speakeasy library is used to support the implementation of Multi-Factor Authentication based on the Time-Based One-Time Password algorithm. Additionally, Axios or Fetch is used to support HTTP request functionality between the frontend and backend.

4.2.4 APIs

The system includes the IP Geolocation API to support the determination of the geographic location of the user based on the user's IP address. This supports the identification of suspicious login activity from unusual locations.

4.2.5 Database Structure

The system uses MongoDB, a NoSQL database, to support the storage of data. The database includes several records, such as administrator records, login logs, calculated risk scores, security alerts, and blocked IP addresses. The records are stored to support the monitoring of login activity to identify suspicious activity.

4.2.6 Algorithms

The proposed system utilizes a number of algorithms to ensure safe authentication and threat detection.

- Password Authentication Algorithm

The proposed system authenticates users based on the password provided by the administrator. The password is compared to the hashed password stored in the database.

- JWT Authentication Algorithm

JSON Web Token is used to generate tokens to ensure safe authentication. The token is used to manage sessions and authenticate users.

- TOTP Multi-Factor Authentication Algorithm

The proposed system utilizes a Time-Based One-Time Password algorithm to generate a token. The administrator has to provide the token to authenticate themselves.

- Rule-Based Suspicious Activity Detection Algorithm

The proposed system utilizes a number of rules to detect suspicious activities. The rules are based on login history. Suspicious activities detected by the proposed system include repeated login attempts, unusual login frequency, or login attempts from unusual locations.

The proposed system utilizes a weighted risk score algorithm to calculate the risk associated with a particular activity. The risk score is based on a number of security events.

- Automatic IP Blocking Algorithm

If the risk score is greater than a predetermined threshold, the proposed system blocks the IP address. The proposed system sends a security alert to administrators.

Chapter 5

Results and Discussion

5.1 Results

The proposed Security Operations Dashboard for Threat Monitoring and Risk Analysis has been successfully implemented to monitor login activities and detect suspicious behavior through a behavior-based risk scoring model. The system continuously records login activities and analyzes them through predefined security rules to detect abnormal behavior that could potentially lead to security threats.

5.1.1 Security Operations Dashboard

The above figure presents the Security Operations Dashboard, where it can display real-time information about login activities, risk scores, security alerts, and blocked IP addresses.

5.1.2 Login Event Monitoring

The system records login activities, storing the following information about the login attempts in the database:

IP address of the user	Timestamp of the login attempt	Status of the login attempt	Geographic location of the login request
------------------------	--------------------------------	-----------------------------	--

These login activities are analyzed to detect suspicious behavior, such as repeated login attempts, login attempts from new locations, or login attempts at unusual frequencies.

5.1.3 Risk Score Evaluation

The system uses the risk score to evaluate the security signals. Each security signal is assigned a weight value to the overall risk score.

The risk score evaluation mechanism is depicted in the table below:

Table 5.1: Risk Score Mechanism

Security Event	Risk Score
Failed login attempt	+10
Login attempt from new location	+20
Multiple login attempts	+30

The system uses the overall risk score to evaluate the security threats. The system classifies the security threats based on the threat confidence level.

Table 5.2: Threat Confidence Level

Risk Score Range	Threat Level
Risk Score < 20	Low Risk
Risk Score 20 – 40	Medium Risk
Risk Score > 40	High Risk

If the risk score is greater than or equal to the threshold, then the IP address is automatically blocked, and a security alert is triggered.

5.1.4 Threat Detection Example

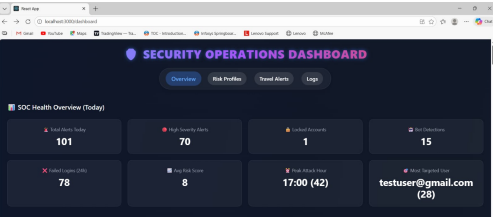
The following are some examples of how the system can detect potential threats:

- Normal login activity: If a login request is made by an IP address that is already registered with valid login credentials, then the risk score is low, which is normal activity.
- Suspicious login activity: If multiple failed login attempts are made by an IP address, then the risk score is increased, which is considered suspicious activity.
- High risk activity: If an unknown location is detected with multiple failed login attempts, then the risk score is greater than or equal to the threshold, which is considered a high risk activity. The IP address is automatically blocked.

5.1.5 False Positive Reduction

To avoid false positives, a risk decay feature is used. If no suspicious activity is detected for a specific period of time, then the risk score is decreased to normal.

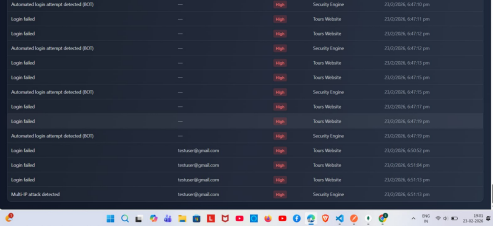
graphicx subcaption



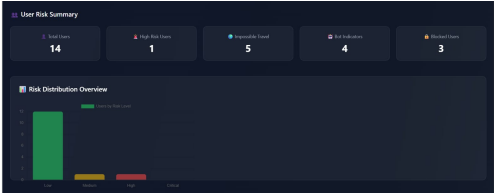
(a) Security Operations Dashboard Overview

User ID	Risk Score	Risk Level	Status	Signals	MITRE	Last updated
test-user-1234	65	High	Active	Multiple failed logins, Impossible travel detected	T1028	2023/05/10 03:57 pm
test-user	50	Medium	Locked	Missing user agent header, Missing User-Agent, Unusually fast login attempts, Multiple failed login attempts	—	2023/05/09 03:57 pm
user123	15	Low	Active	Missing user agent header, Missing User-Agent, Multiple failed login attempts, Impossible travel detected	—	2023/05/09 03:57 pm
user123	0	Low	Active	Multiple failed login attempts, Impossible travel detected	T1028, T1029	2023/05/10 03:57 pm
test-user@domain.com	0	Low	Active	Unusually fast login attempts, Multiple failed login attempts	—	2023/05/09 03:58 pm
test-user-1	0	Low	Active	Missing user agent header, Missing User-Agent, Unusually fast login attempts	—	2023/05/10 03:58 pm
testuser	0	Low	Active	Impossible travel detected	T1028	2023/05/10 03:58 pm

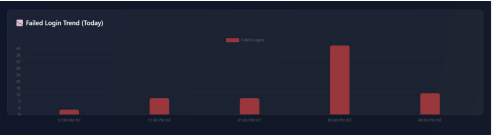
(b) User Risk Profile Table



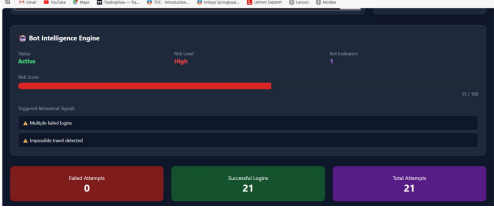
(c) Security Event Log Monitoring



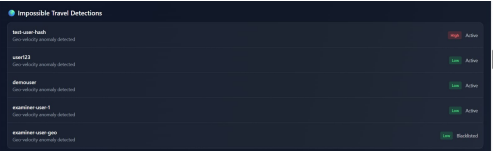
(d) User Risk Summary Dashboard



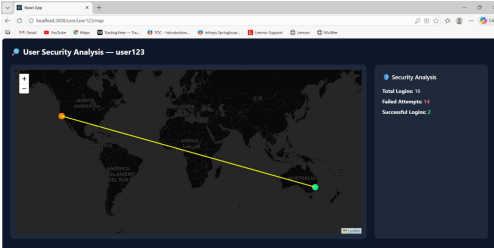
Failed Login Trend Dashboard



(e) Bot Intelligence Engine – High Risk Detection



(f) Impossible Travel Detection Alerts



(g) User Security Analysis Map

Figure 5.1: System Screenshots

5.2 Discussion

The results have shown that the proposed system is able to monitor and detect login activities effectively using a risk analysis approach. Unlike other machine learning-based systems, the proposed system is able to provide explainable threat detection, where each suspicious activity is contributing to the final risk score.

The system is also able to combine multiple behavioral indicators, such as login location, frequency, and failed attempts, to minimize the chances of false positives and ensure reliable threat detection. The Security Operations Dashboard is able to provide real-time visibility of system security events to help administrators detect potential threats and take necessary security actions.

Chapter 6

Conclusion

The proposed Security Operations Dashboard for Threat Monitoring and Risk Analysis was created to monitor login activities, detect unusual behavior, and identify potential security threats. The proposed system uses a number of security mechanisms, such as secure login authentication, Multi-Factor Authentication, IP verification, login event recording, and risk score analysis**. By analyzing login activities, the proposed system was able to detect unusual behavior, such as repeated login attempts and login attempts from unusual locations.

The proposed system was able to evaluate the risk level of the unusual behavior through the implementation of a risk-based scoring system. This system was able to block malicious IP addresses if the risk score exceeded a certain value. The proposed system was able to provide a dashboard where administrators could easily monitor login activities, security threats, and blocked users. Therefore, the proposed system was a good example of a cost-efficient way of improving the security of a system against unauthorized access.

Publications by the candidates

International Journals

1. Nitin Shivsharan, Sanjay Ganorkar "Diabetic Retinopathy Detection using Optimization Assisted Deep Learning Model: Outlook on Improved Grey Wolf Algorithm" International Journal of Image and Graphics vol. 21 issue 1 Jan .2021 ISSN NO 0219-4678.[[Web of Science and Scopus Indexed](#)] [Singapore](#).
2. Nitin Shivsharan, Sanjay Ganorkar. "Automated Diabetic Retinopathy Detection using Convolutional Neural Network". International Journal of TEST Engineering and Management, vol. 83 issue March April: 2139–2145, April 2020. ISSN NO 0193-4120.[[Scopus Indexed](#)] [USA](#).

International Conferences

1. Nitin Shivsharan, Sanjay Ganorkar. "Identification and Prediction of Chronic Diseases Using Deep Learning Models". IEEE Conferences on Computer Vision and Pattern Recognition (CVPR 2022). [Scopus Indexed](#) [New Orleans, USA](#). [Accepted]
2. Nitin Shivsharan and Sanjay Ganorkar "Implementation of Modified Pre-trained DenseNet Model for the Classification of Grades of the Diabetic Retinopathy" Springer's 5th International Conference on Inventive Communication and Computational Technologies ICICCT 2021 25-26

June, Springer Lecture Notes in Networks and Systems
Scopus Indexed Chennai.

Implementation Plan

Table 6.1: Implementation Plan

Action plan	2023						2024			
	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr
1. Study the theoretical background and review existing literature										
2. Prepare and present Synopsis report										
3. Setup experiment and start experimental work										
4. Analysis of the experimental results										
5. Prepare the report and presentation										

Table 6.1 presents the implementation plan.

Appendix

Detailed information, lengthy derivations, raw experimental observations, etc. are to be presented in separate appendices, which shall be numbered in Roman Capitals (e.g. “Appendix I”). Since reference can be drawn to published/unpublished literature in the appendices these should precede the “Literature Cited” section.

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